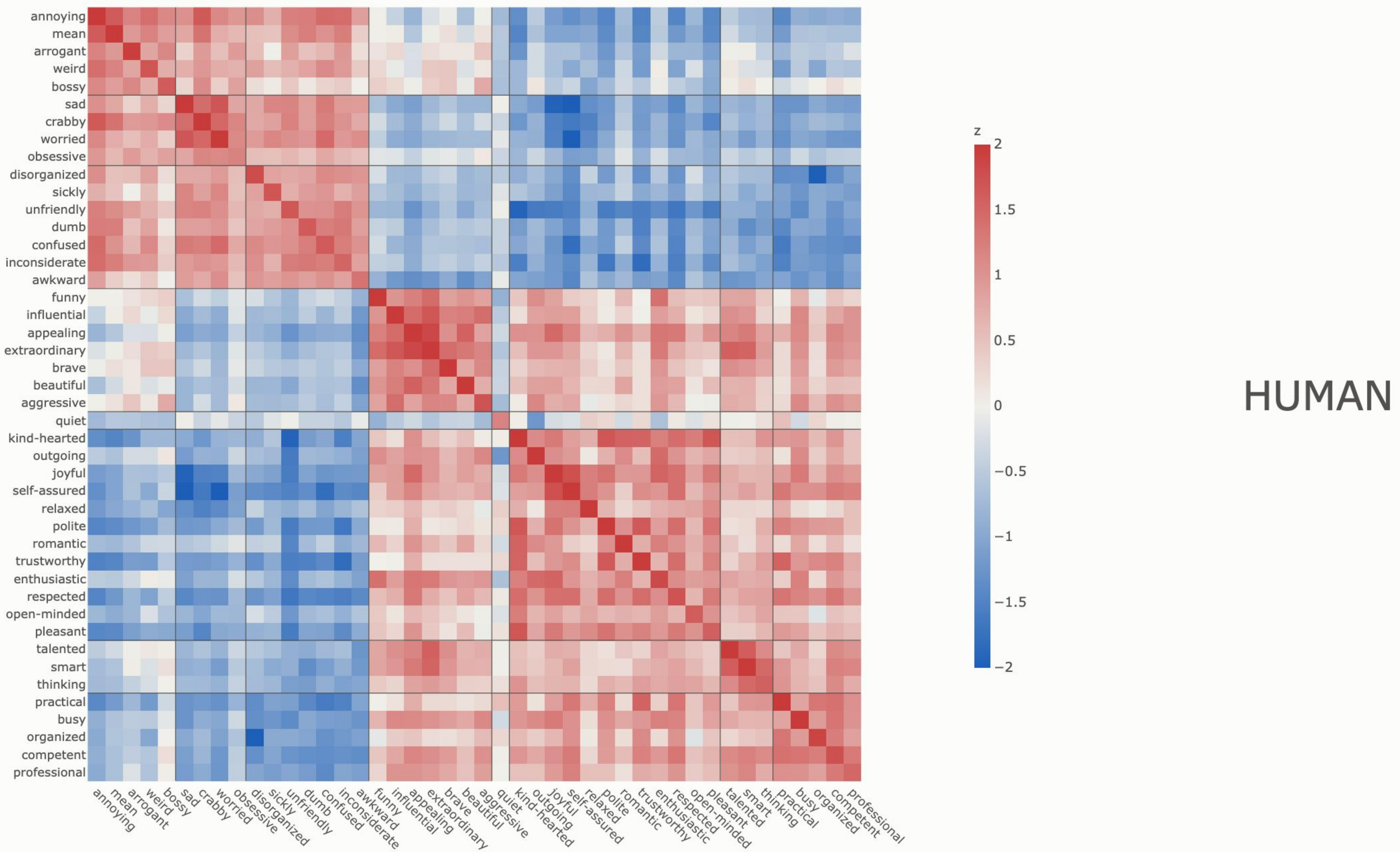
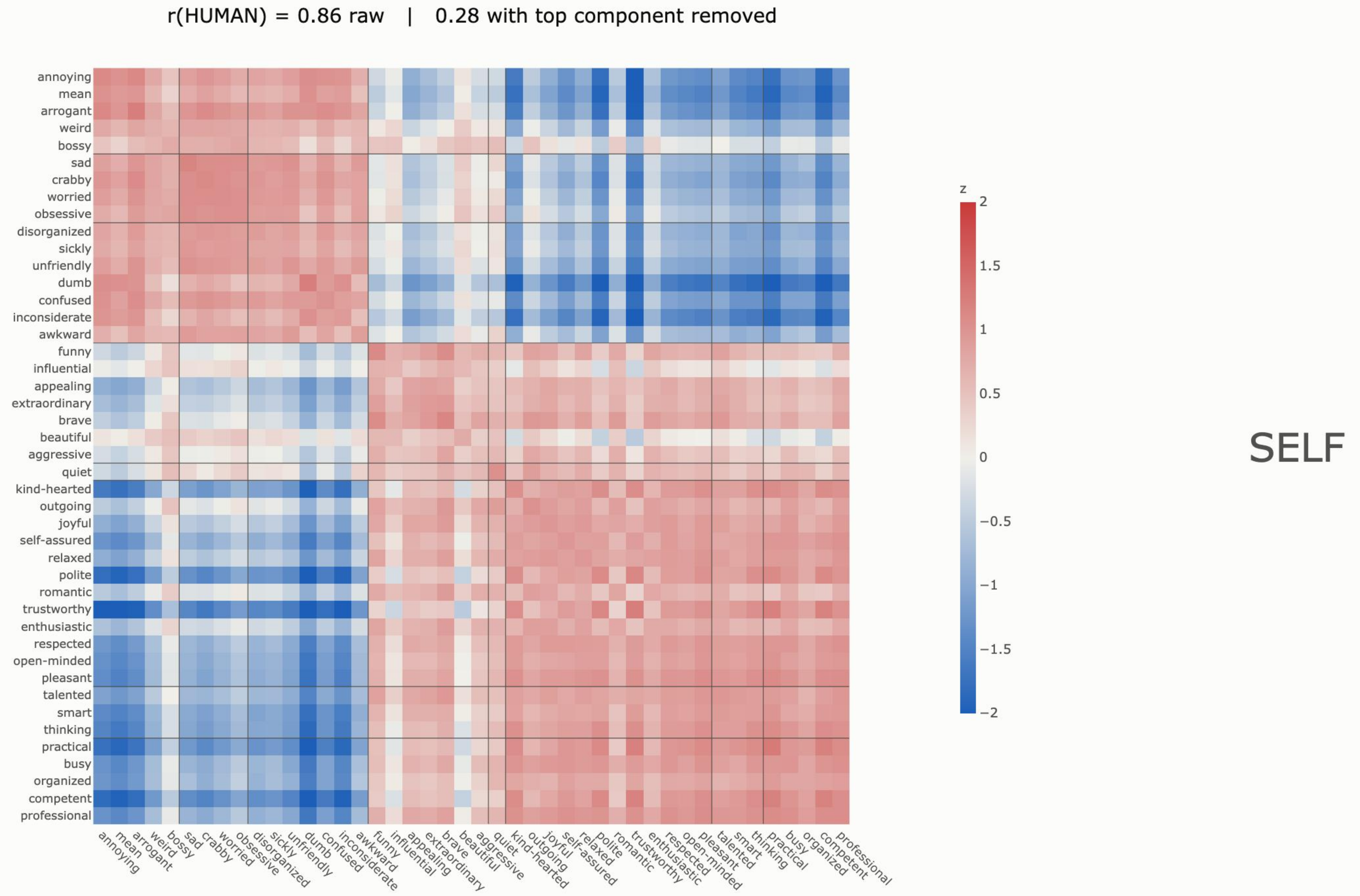
Measured Human Self-Report Correlations

Correlations from Saucier's 525-PDA. For visualization, these are clustered by a signed version of Ward's minimum variance (so antonyms stay separate). Cell value is the mean of the pairwise correlations, with cluster cohesion on the diagonal.



The Same for Models

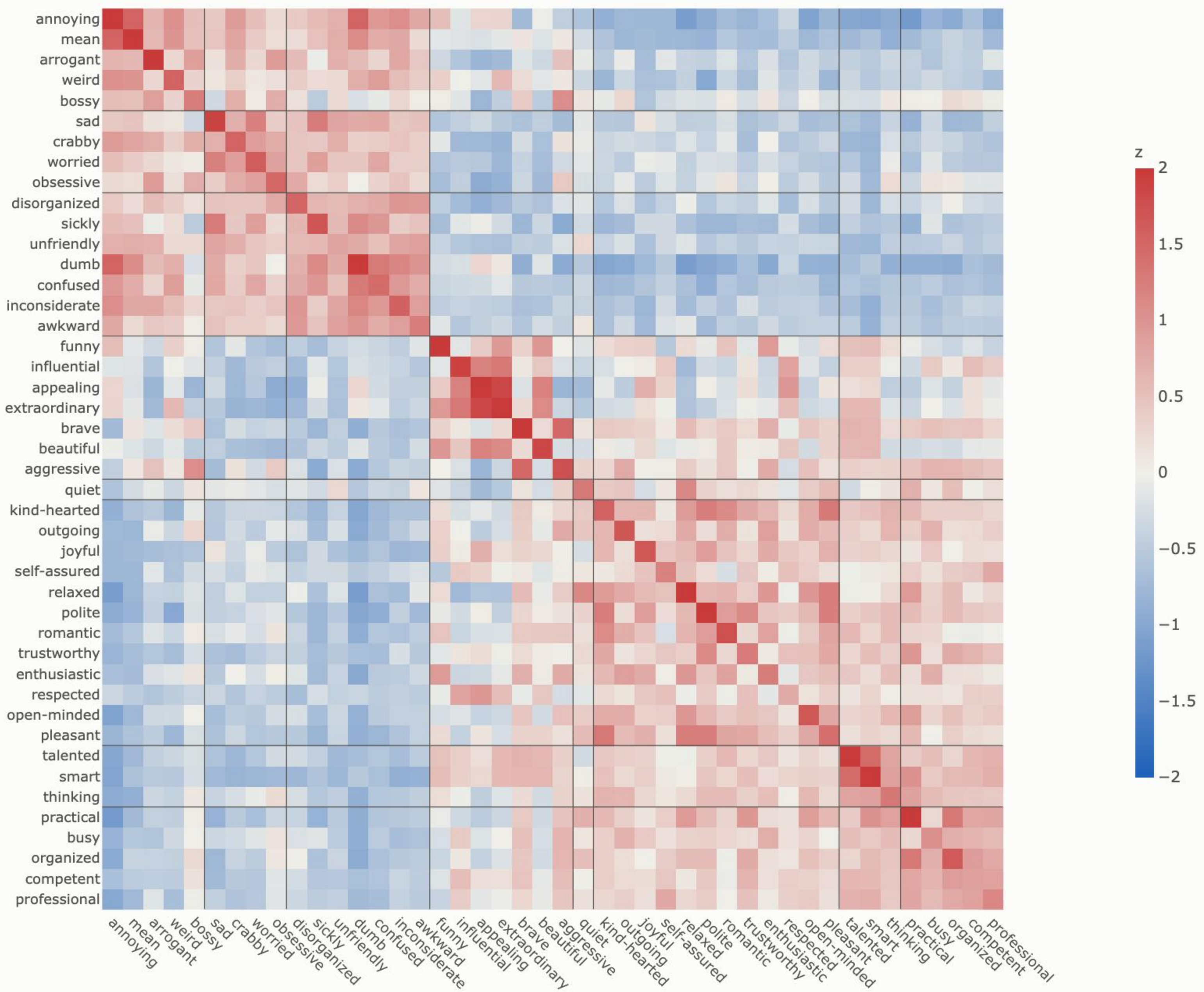
n = 65 small instruct models, mean value across 6 question framings. Correlations drop the scale here, but model's absolute variance is much smaller. It's also easy to see the desirability block washes out everything else from this view.



Models' Residual-stream Representations are More Complex

Gathered as midlayer residual-stream states on adjective input tokens in a personality frame, recentered on all adjectives. Mean cosine similarity across attributes in the clusters across models is shown below. Note the strong diagonal (cluster cohesion), but the off-diagonal quite different from human.

r(HUMAN) = 0.80 raw | 0.41 with top component removed

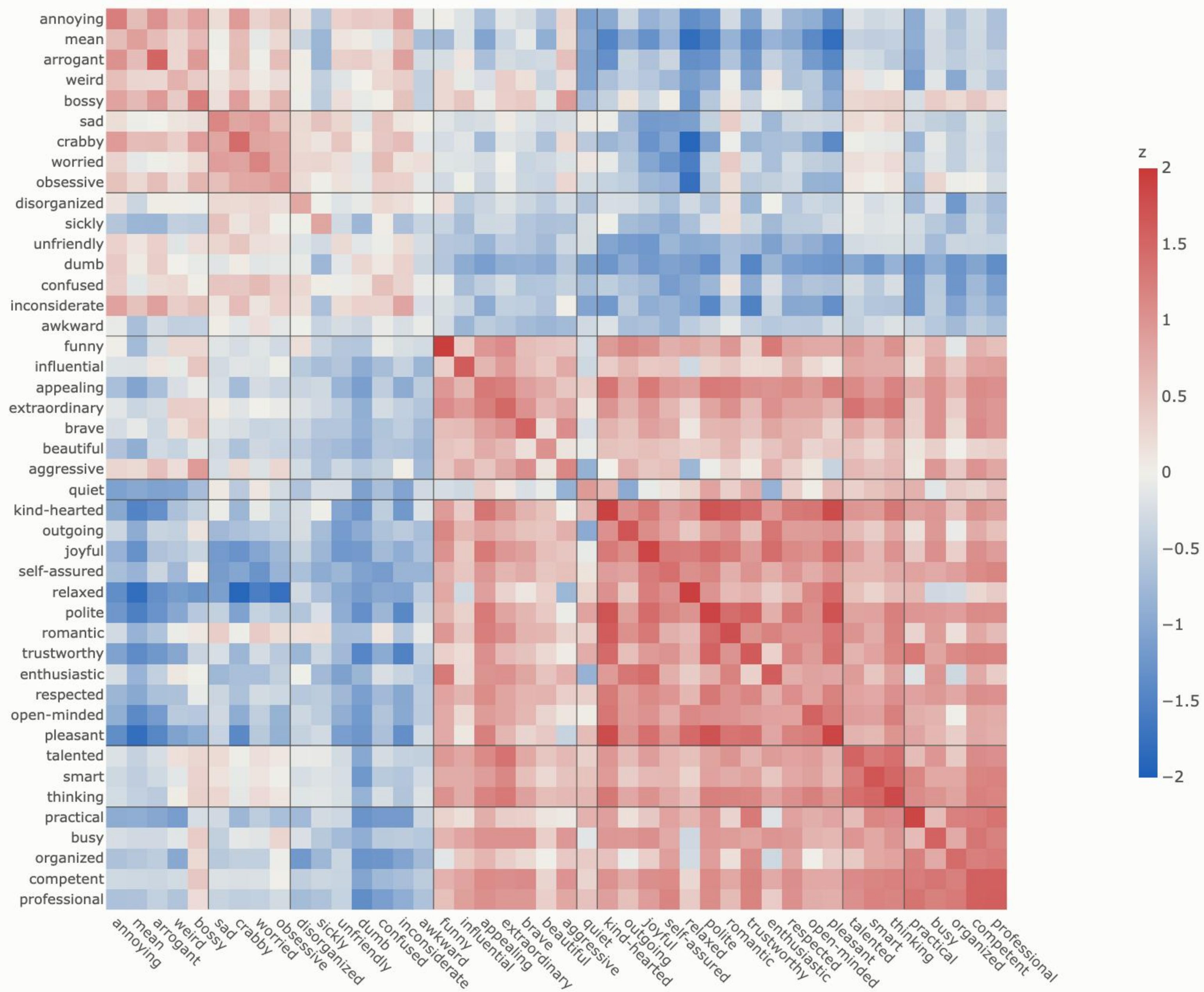


REPRESENT

Asking the Models for Trait Relations

We directly ask the models "How likely is a very {X} person to be {Y}?" And the models step up: this is the best match of human behavior. Here we zscore the entries and symmetrize.

r(HUMAN) = 0.86 raw | 0.77 with top component removed

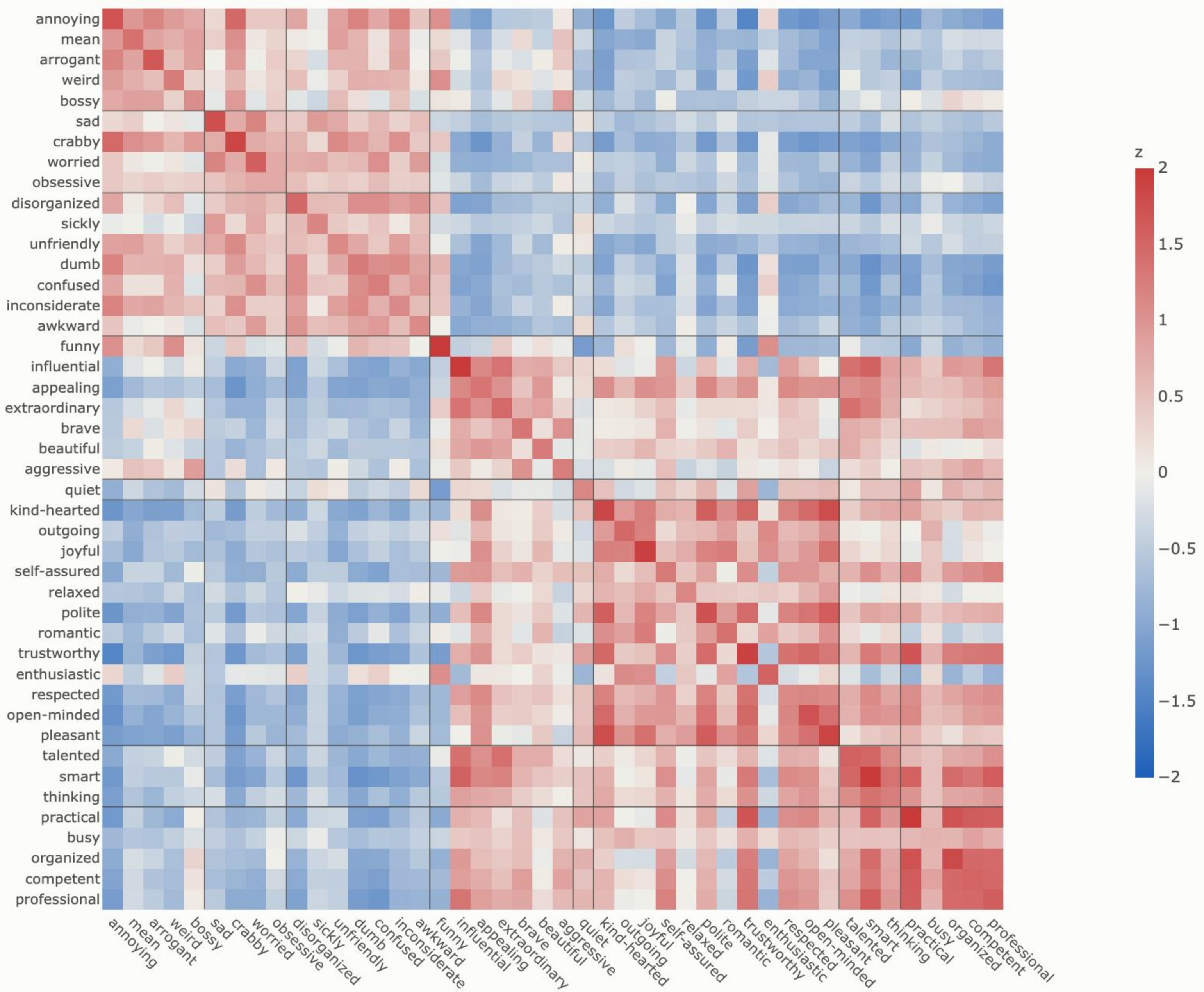


JUDGE

But, Tasked to Enact this Understanding, Models Fall Short

This time we ask models to enact the given attributes when answering user questions. We again gather mid-layer activations, and recenter across attributes, to come up with enacted-persona vectors. Again, cosine similarity, meaned across models, is shown below. These models, at least, seem to have trouble being funny without being annoying.

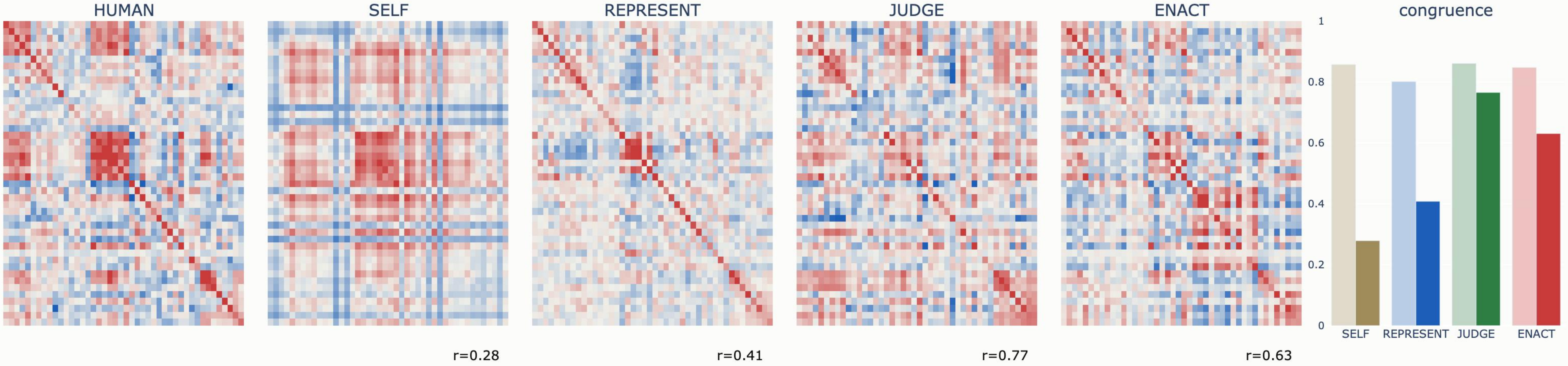
r(HUMAN) = 0.85 raw | 0.63 with top component removed



ENACT

Correspondence with Humans is Often Shallow

If you remove each instrument's top principal component, the visual shapes become notably different, and the congruence with human falls notably, except for JUDGE.



Congruence bars: faded = full, solid = PC1 removed.